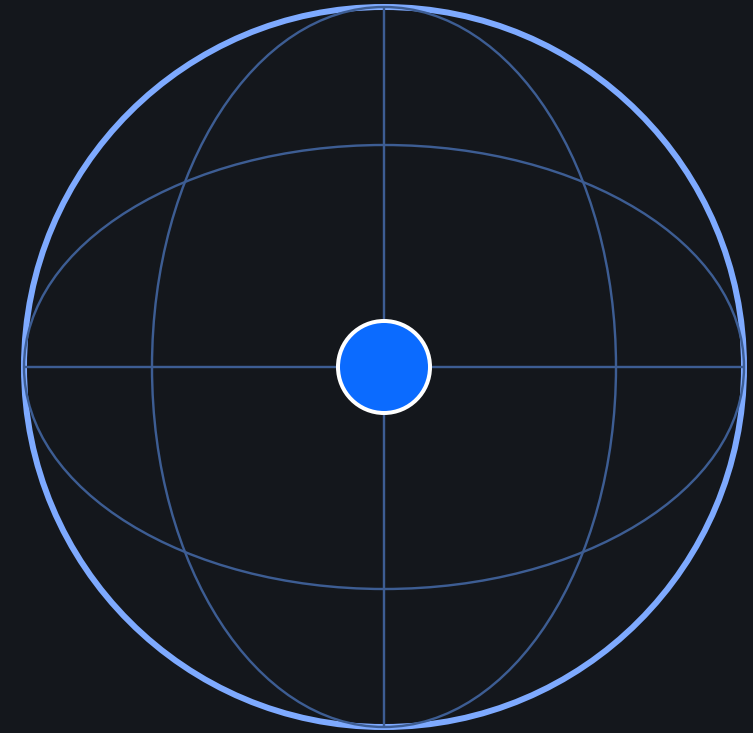


SPHERES

What exists — and what to build next

A visual map of the simulation, player-facing systems, open gaps, and the next milestone that connects the whole game.



MAP

SIMULATION

PLAYER

The next step is not another isolated system. It is a loop that makes the existing systems depend on one another.

The game already has a broad playable backbone

137

nations at the 1990 start

328

technologies across 8 domains

12

separate resource lines

World simulation

Deterministic Rust core, daily player clock, monthly system settlement, saves, state hashes, headless runs, and calibration tests.

Player agency

Economy, annual budget, research priorities, diplomacy, government, resource deals, procurement, and a nine-rung conflict ladder.

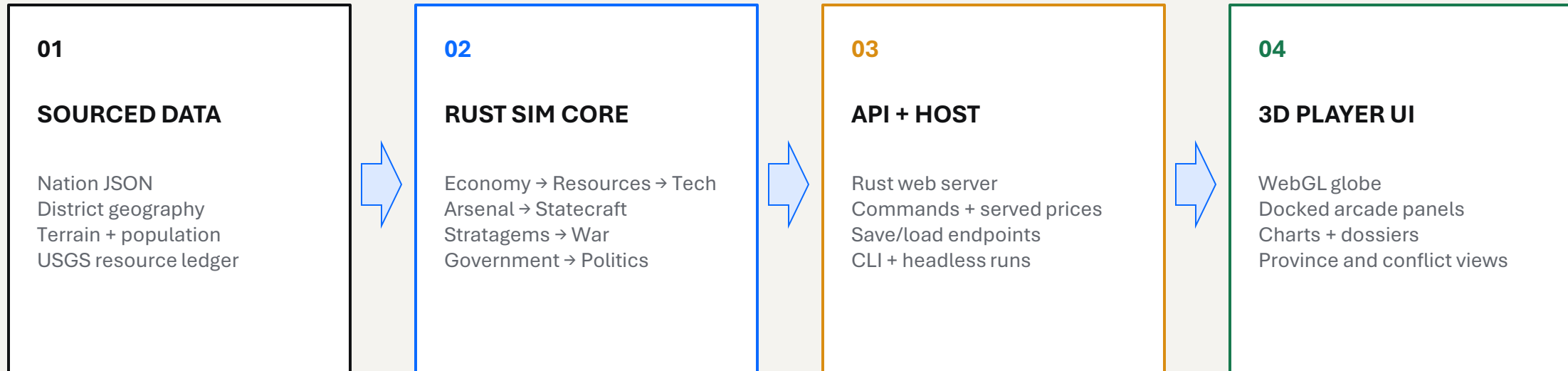
Visible world

Full 3D globe, province selection, changing province population, terrain, resources, cities, contracts, fronts, history, and dossiers.

BOTTOM LINE

The foundation is wide enough. The priority now is to make the systems form one repeatable player loop.

Four layers turn sourced world data into player decisions

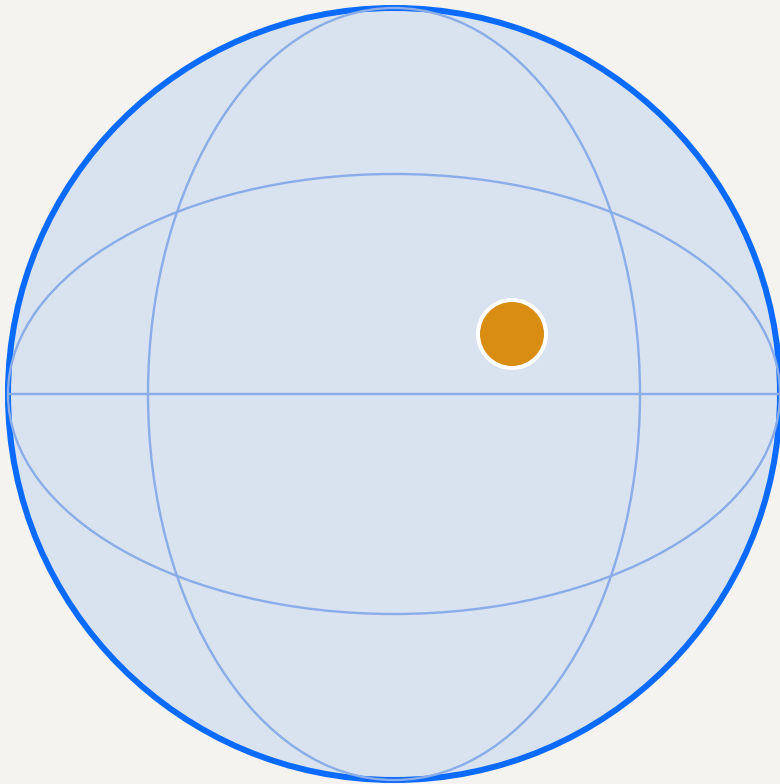


DETERMINISTIC TIMELINE

Commands are applied first; monthly systems run in a fixed order; saves serialize the full world; a state hash proves two runs agree everywhere.

Current delivery: library + CLI + browser server. Native desktop packaging is not yet the product layer.

The globe is now the primary strategic surface



CLICK + DRAG

ZOOM TO CITIES

01 Province/state selection

Click an admin-1 district to lock a dossier with owner, population, terrain, resource holdings, and local context.

02 Living province population

1990 district residents evolve with the current owner's demographic and technology path, including after border transfers.

03 Strategic overlays

Political ownership, terrain and relief, resources, contracts, aims, rivers, cities, labels, and war/front information.

04 Daily player calendar

The displayed date advances day by day; calibrated GDP, research, procurement, war, and politics still settle monthly.

PLAYABLE

Economic policy is a full yearly allocation, not a single boost button

Macro model

PLAYABLE

GDP growth and convergence, inflation, interest rates, taxes, debt, employment, investment, oil terms, sanctions, bubbles, crises, and war drag.

Annual settlement

PLAYABLE

Ten ministry allocations are voted as one fiscal plan. Policy changes consume political capital and feed multiple downstream systems.

PLAYER READ

Revenue → allocations → deficit/debt → growth and political consequences

THE 10 BUDGET DIALS

Health

health + population

Education

skills + research

Families

demography + stability

Pensions

stability + fiscal load

Infrastructure

investment efficiency

Industry

productive capacity

Science

research output

Defense

force + procurement

Security

regime resilience

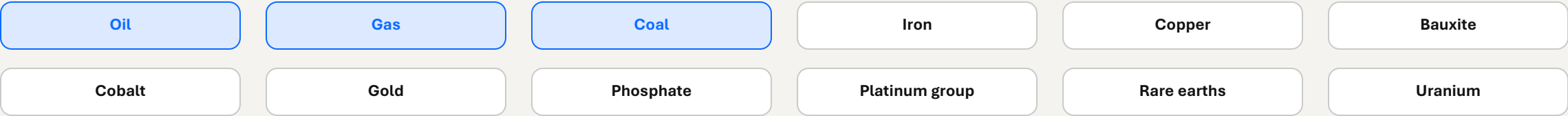
Diplomacy

statecraft reach

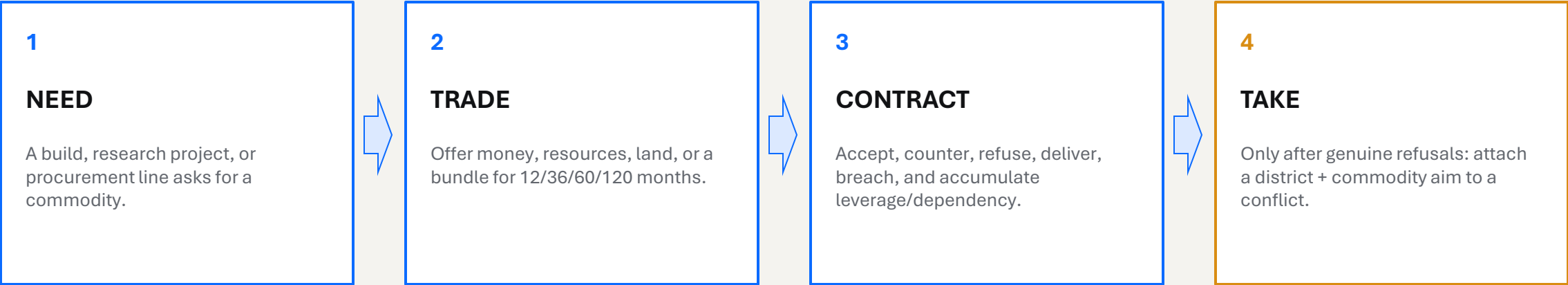
The system exists; the next design task is making tradeoffs immediately legible and fun at arcade speed.

Resources and trade connect provinces, production, diplomacy, and war

12 SEPARATE LINES



HAVE · STOCK · NEED · action cover



INTEGRATED Open: MINE/development gameplay, consumers for six newer lines, and tuning the refusal chain so last-resort war is rare but reachable.

Political capital ties diplomacy and domestic authority together

POLITICAL CAPITAL

Every major action has a served price. Governments can run out of authority, while reneging remains possible but reputationally ruinous.

EXTERNAL POWER

- 01 Improve relations · sanctions
- 02 Mutual-defense pacts · guarantees
- 03 Aid and patronage transfers
- 04 Trade depth · dependency · leverage
- 05 Covert action and discovery
- 06 Access and basing for theatres

DOMESTIC POWER

- 01 Parties, parliaments, cabinets, and coalitions
- 02 Elections and early elections
- 03 Regimes and institutional pillars
- 04 Stability, separatism, and collapse
- 05 Policy pricing and cabinet constraints
- 06 Stratagem decision cards

PLAYABLE

War is an escalation decision before it becomes an invasion



QUARREL / PRESSURE

SHOOTING WAR → TERRITORY

Commitment

Each participant chooses its own rung, objective, rules of engagement, ceiling, and resolve red line.

Access + reach

Theatres, sustaining bases, permissions, deployable fractions, and small-state leverage determine what can arrive.

Operational front

District terrain, rivers, population, force packages, magazines, tempo, exposure, and encirclement shape combat.

Settlement

Resolve and exhaustion lead to ceasefires, concessions, subjugation, annexation, or resource-targeted district transfer.

Nuclear taboo prevents direct wars between nuclear powers; escalation and interventions can still reshape third-party conflicts.

PLAYABLE

Technology creates long-term capability; the arsenal turns it into force

8 RESEARCH DOMAINS



Research is funded by output, directed by a national priority and named focus, constrained by prerequisites and year floors, and accelerated by diffusion from open trade partners.

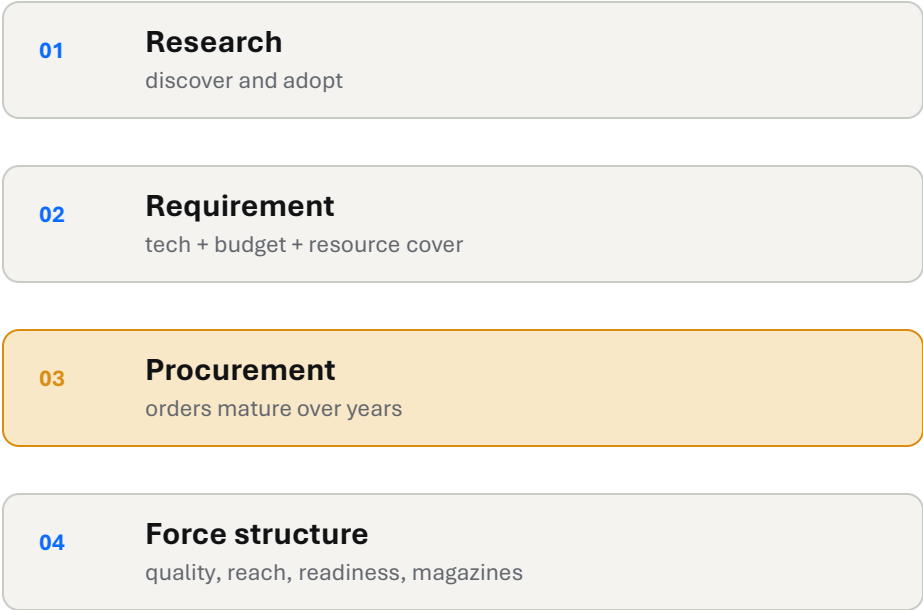
328

technology entries

40y

hardware service life can span decades

CAPABILITY PIPELINE



INTEGRATED

Open: add the remaining uranium / rare-earth / battery / hydrogen consumers and expose blocked requirements cleanly in the UI.

The interface is moving toward arcade-style progressive disclosure

SURFACE	WHAT EXISTS	STATUS
Base screen	3D globe + date + only high-value readouts	PLAYABLE
Context on demand	Province dossier, nation sheet, conflicts, tech, economy, trade, resources	PLAYABLE
Arcade interaction	Large cards, resource chips, served costs, clear blocked reasons, folded advanced detail	INTEGRATED
Persistence	JSON save/load, deterministic seeds, replayable headless timeline	PLAYABLE
Delivery today	Rust library + CLI + localhost browser server	DEVELOPER-READY
Standalone product	Desktop wrapper, installer, settings, save slots, autosave, crash recovery	PLANNED

PRODUCT RULING

Prove the core loop first; package it as a standalone app immediately after the loop is worth replaying.

Most foundations are playable; the missing work is concentrated

SYSTEM	STATUS	NEXT GAP
World map + provinces	PLAYABLE	Globe, dossiers, terrain, population, city/resource overlays
Economy + annual budget	PLAYABLE	Complete model and 10 levers; needs sharper arcade feedback
Resources + stockpiles	INTEGRATED	12 lines and action gates; development and six consumers remain
Negotiated trade	INTEGRATED	Bundles/contracts work; market loop and war reachability need tuning
Statecraft + government	PLAYABLE	PC-priced external and domestic action systems
Conflict + war	PLAYABLE	Ladder, access, fronts, terrain, settlement; calibration debt open
Technology + arsenal	INTEGRATED	328-entry tree and procurement; more resource gates remain
Standalone delivery	PLANNED	Core is native Rust; desktop product packaging is not built
CONCENTRATION OF WORK: resource development + consumers + market tuning + campaign feedback		

Next milestone: complete the resource-security player loop

WHY THIS NEXT **It connects the province map → resources → economy and budget → trade and statecraft → procurement → war, using systems that already exist.**

01

DEVELOP

Build the MINE card: district project, money/PC cost, lead time, capacity, yield, and ownership risk.

The map becomes economically actionable.

02

CONSUME

Add honest requirements for uranium, rare earths, batteries, hydrogen, magnets, reactors, and the bomb.

Stockpiles create meaningful choices without ambient GDP punishment.

03

TRADE + TUNE

Complete AI offers/counters, market temperature, sanctions/rationing, refusal memory, and last-resort-war rates.

The player can solve scarcity without war — until the world closes every door.

04

REWARD

Add yearly resource-security scorecards, alerts, goals, and event-style payoffs for resilient supply chains.

The loop has readable wins, losses, and replay value.

DEFER **New commodity types until the 12-line loop is fun and calibrated.**

THEN: standalone desktop packaging